ApontaUA

FUNCTIONAL REQUIREMENTS

Collaborative Academic Document Management System

FUNCTIONAL REQUIREMENTS | WORKING DRAFT

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Document Control

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1. Purpose and Scope

This document defines proposed functional requirements for ApontaUA, a collaborative platform for publishing, organizing, searching, reviewing and reusing academic materials produced by students and teachers at the University of Aveiro. The requirements reflect the team's agreed high-level use cases and current discussions. This is a working baseline, not a claim that every feature has been approved or implemented.

1.1 Actors

- Student — searches, publishes and comments on academic materials.
- Teacher — publishes official materials and validates academic materials within the scope of their permissions.
- Course Unit Responsible Teacher — defines validation criteria and delegates validation permissions for the relevant course unit.
- Administrator — manages users, reports, moderation, academic structure, platform rules and audit records.

1.2 Priority Definitions

Must — required for the agreed MVP scope. Should — important but may be deferred. Could — optional enhancement. Priorities are provisional and must be confirmed by the team and supervisors.

1.3 Supported MVP File Formats

Proposed formats: PNG, JPG/JPEG, PDF, DOCX, PPTX, Markdown (.md) and Excel (.xlsx). The original file should be preserved. File acceptance, preview, text extraction, search and in-platform editing are separate capabilities; not every format supports all of them. Legacy Excel .xls is excluded unless explicitly approved.

2. Authentication and Access

ID	Requirement	Priority
FR-001	The system shall allow users to sign in through an authentication mechanism supported by the platform.	**Must**
FR-002	The system shall identify each authenticated user's role and enforce role-based permissions.	**Must**
FR-003	The system shall support at least the Student, Teacher and Administrator roles.	**Must**
FR-004	The system shall allow an administrator to suspend and reactivate user accounts.	**Must**
FR-005	The system shall prevent users from performing actions for which they do not have permission.	**Must**

3. Academic Structure and Discovery

ID	Requirement	Priority
FR-006	The system shall organize academic materials by course, course unit and topic.	**Must**

ID	Requirement	Priority
FR-007	The system shall allow users to browse materials associated with a course unit or topic.	Must
FR-008	The system shall allow users to search materials using keywords.	Must
FR-009	The system should allow users to filter results by course unit, topic and material type.	Should
FR-010	Search results shall display title, author, material type and publication date.	Must
FR-011	The system shall display the academic validation status of a material when applicable.	Must
FR-012	The system should allow users to browse materials from previous academic years.	Should

4. Material Submission and File Handling

ID	Requirement	Priority
FR-013	The system shall allow authenticated students to submit academic materials.	Must
FR-014	The system shall allow teachers to publish official academic materials.	Must
FR-015	The system shall accept PNG, JPG/JPEG, PDF, DOCX, PPTX, Markdown (.md) and XLSX files, and explain when a file is rejected as unsupported.	Must
FR-016	The system shall require a title, course unit, material type and topic metadata where applicable for each material.	Must
FR-017	The system shall record the material's author and publication date.	Must
FR-018	The system shall preserve the original uploaded file without modifying it during text extraction or conversion.	Must
FR-019	The system shall allow authorized users to view or download the original file.	Must
FR-020	The system should provide an in-platform preview when technically supported for the format.	Should
FR-021	The system shall allow editing of materials created in formats supported by the platform editor.	Must
FR-022	The system should allow an uploaded file to be replaced by a newer version while preserving previous versions.	Should
FR-023	The system shall report file-processing failures without silently discarding an otherwise accepted original file.	Must

5. OCR and Text Extraction

ID	Requirement	Priority
FR-024	The system shall attempt to extract text from submitted PNG, JPG and JPEG images using OCR.	Must
FR-025	The system shall attempt OCR on scanned PDFs or PDF pages that contain images rather than extractable text.	Must
FR-026	The system shall identify text segments whose OCR confidence is below a configurable threshold, when confidence information is available.	Must

ID	Requirement	Priority
FR-027	The system shall highlight uncertain extracted text and allow the user to inspect the corresponding region of the original image or page.	Must
FR-028	The system shall allow users to correct or manually transcribe text that was extracted incorrectly or could not be recognized.	Must
FR-029	For uncertain segments, the system shall ask the user to confirm or correct the text and may suggest uploading a clearer image with more legible handwriting.	Must
FR-030	The system shall distinguish automatically extracted text from text confirmed or corrected by the user.	Must
FR-031	The system shall preserve the original image or document regardless of OCR results or corrections.	Must
FR-032	The system shall make usable extracted or user-confirmed text available for search and similarity detection.	Must
FR-033	The system shall attempt to extract text from PDF, DOCX, PPTX, XLSX and Markdown files when technically possible.	Must
FR-034	The system should indicate whether text extraction succeeded, failed or produced limited results.	Should

6. Duplicate and Similarity Detection

ID	Requirement	Priority
FR-035	The system shall perform a duplicate or similarity check for every material submission, regardless of supported file format.	Must
FR-036	The system shall identify exact file duplicates using a suitable file fingerprint or hash.	Must
FR-037	When usable text is available, the system shall compare normalized text to identify potentially similar materials.	Must
FR-038	The system should use MinHash with banded Locality-Sensitive Hashing (LSH) to retrieve candidate documents with similar text.	Should
FR-039	The system shall present potentially duplicate or similar materials to the submitter before finalizing publication.	Must
FR-040	The system shall allow the submitter to inspect candidate materials before deciding how to proceed.	Must
FR-041	The system shall not automatically reject a material solely because it is similar to another material.	Must
FR-042	The system shall allow distinct materials on the same topic to coexist when they serve different purposes or contain meaningfully different content.	Must
FR-043	When text extraction is unavailable, the system shall still attempt exact-file duplicate detection and shall not imply that text similarity was checked successfully.	Must

7. Content Moderation and Reports

ID	Requirement	Priority

ID	Requirement	Priority
FR-044	Before making submitted content publicly available, the system shall run the configured content-moderation checks.	Must
FR-045	The moderation process shall return one of three outcomes: Approved, Pending Review or Blocked.	Must
FR-046	The system shall route content for human review when the automated assessment is ambiguous or insufficiently confident.	Must
FR-047	The system shall prevent publication when content meets a clearly defined blocking rule.	Must
FR-048	The system shall inform the submitter when content is blocked or sent for review and provide an appropriate explanation.	Must
FR-049	The system shall allow administrators to inspect content awaiting moderation review.	Must
FR-050	The system shall allow users to report materials they believe violate platform rules.	Must
FR-051	The system shall allow administrators to review reports and decide whether content should remain available, be hidden or be removed.	Must
FR-052	The system shall record moderation decisions, timestamps and the responsible actor for audit purposes.	Must
FR-053	The system shall keep content moderation separate from academic validation; passing one process shall not imply passing the other.	Must
FR-054	The system should account for academic context and avoid blocking content solely because it contains quoted offensive language, technical terms or controversial subject matter.	Should

8. Academic Validation and Delegation

ID	Requirement	Priority
FR-055	The system shall allow the responsible teacher for a course unit to validate materials associated with that course unit.	Must
FR-056	The system shall allow the responsible teacher to delegate validation permissions to other teachers who teach the same course unit.	Must
FR-057	The system shall allow the responsible teacher to view current validation delegates for the course unit.	Must
FR-058	The system shall allow the responsible teacher to revoke a previously granted validation delegation.	Must
FR-059	The system shall scope each delegated permission to the relevant course unit and shall not grant access to other course units by default.	Must
FR-060	The system shall allow authorized teachers to approve or reject materials submitted for academic validation.	Must
FR-061	The system should allow the responsible teacher to define validation criteria for their course unit.	Should
FR-062	The system shall notify the material's author when its validation status changes.	Must
FR-063	The system shall allow an authorized teacher to revoke a previous validation within the scope of their permissions.	Must
FR-064	The system shall display the material's current academic validation status to authorized viewers.	Must

ID	Requirement	Priority
FR-065	The system shall record who approved, rejected or revoked validation and when the action occurred.	Must

9. Comments and Interaction

ID	Requirement	Priority
FR-066	The system shall allow authorized users to post comments associated with academic materials.	Must
FR-067	The system shall display comments to users authorized to view the corresponding material.	Must
FR-068	The system should allow users to report comments they consider inappropriate.	Should
FR-069	The system shall allow administrators to moderate comments that violate platform rules.	Must

10. Version History

ID	Requirement	Priority
FR-070	The system shall create a new version when an editable material is changed and saved.	Must
FR-071	The system shall retain previous versions of each material.	Must
FR-072	The system shall allow authorized users to view a material's version history.	Must
FR-073	The system shall record the author and timestamp of each version.	Must
FR-074	The system should record the validation status associated with each version, where applicable.	Should
FR-075	The system shall preserve previous versions when an uploaded file is replaced.	Must
FR-076	The system should require a changed material to undergo the applicable validation workflow again rather than automatically carrying over the previous version's validation.	Should

11. Administration and Audit

ID	Requirement	Priority
FR-077	The system shall allow administrators to manage general user roles and permissions.	Must
FR-078	The system shall allow administrators to manage the platform's course and course-unit structure.	Must
FR-079	The system should allow administrators to configure general platform rules within their authorized scope.	Should
FR-080	The system shall allow administrators to consult audit logs of relevant system operations.	Must
FR-081	The system shall record relevant administrative actions, including permission changes and moderation decisions.	Must

12. Proposed Moderation Policy (To Be Validated)

This is a starting proposal for discussion, not a finalized policy. Automated moderation should be conservative and context-aware. The system should distinguish an automated assessment from a human moderator's final decision.

Outcome	Proposed behaviour
Approved	No meaningful evidence of a rule violation is detected. Publication may proceed, subject to duplicate detection and other workflows.
Pending Review	The content is ambiguous, confidence is insufficient, context is missing, or OCR quality prevents a reliable assessment. Keep it unavailable publicly until reviewed.
Blocked	A clearly defined prohibited category or technical/security rule is matched with sufficiently strong evidence. Prevent publication and explain the reason to the submitter.

12.1 Suggested Initial Rule Categories

- Block only clear, high-confidence violations of explicitly defined platform rules or file-security restrictions.
- Send potentially abusive, discriminatory or harassing content for human review when context or confidence is uncertain.
- Do not block solely because content contains quoted offensive language in an academic context, technical terminology, controversial historical material or code examples.
- Do not treat academic errors, low quality or similarity to another document as content-moderation violations; route them through academic review or duplicate handling.
- If OCR output is too incomplete to support a reliable moderation decision, request correction or route the submission for review rather than assuming the content is safe or unsafe.

13. Open Decisions

Topic	Decision to confirm
File limits	Set maximum upload size and any per-format limits.
Excel compatibility	Confirm whether .xls is excluded and only .xlsx is supported.
OCR threshold	Select OCR technology and decide how uncertain segments are identified and displayed.
OCR correction	Decide whether confirmation is required for all OCR output or only uncertain segments.
Similarity thresholds	Evaluate file hashes, MinHash/LSH thresholds and image perceptual hashing; measure false positives.
Moderation categories	Approve prohibited categories, escalation rules, reconsideration process and retention of moderation data.
Delegation	Confirm whether delegated teachers may validate their own uploads and whether further delegation is prohibited.
Validation workflow	Define rejection reasons, criteria and what happens to validation after an edit.
MVP boundary	Confirm which Should requirements are included in the MVP.
Privacy and retention	Define access, retention and deletion rules for uploaded materials, OCR text, reports and audit records.

14. Notes on Requirements Quality

- These requirements are a working draft and must be reviewed by the project team and supervisors.
- Must/Should priorities are provisional and do not replace MVP planning.
- Non-functional requirements, including performance, accessibility, security targets and availability, should be specified separately.
- The final specification should map each requirement to one or more agreed use cases and later to acceptance tests.
- MinHash/LSH and OCR are intended capabilities; implementation details and measured thresholds belong in the architecture and technical validation.

15. Project Team and Supervision

Name	Role
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Appendix A — Summary

This draft contains 81 proposed functional requirements. Priorities and the exact MVP boundary must be reviewed by the team and supervisors before the document is baselined.